All 880 Complex Realizations of the Maximal 10-Vertex Laman Graph Can Be Real

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Abstract

A minimally rigid planar graph (or Laman graph) $G = (V, E)$ on n vertices has $2n - 3$ bars and admits a finite number $c_2(G)$ of complex realizations up to direct planar isometries for generic bar lengths. At $n = 10$ vertices, the maximum complex count over all 110,132 Laman graphs is $c_2(10) = 880$, achieved uniquely by the graph L_{880} (OEIS A306420). Whether a real bar-length assignment exists for which all 880 complex realizations of L_{880} are simultaneously real has remained open since the determination of $c_2(10)$ by Capco, Gallet, Grasegger, Koutschan, Lubbes, and Schicho (2018); the sharpest published lower bound due to Bartzos, Emiris, Legerský, and Tsigaridas (2018, 2020) certified 860 real realizations via univariate symbolic root isolation and observed 868 numerically via homotopy continuation. In this paper we give a computer-assisted proof, pending independent verification, that $r_2(L_{880}) = r_2(10) = c_2(10) = 880$. We exhibit 17 explicit positive rational bar lengths for which the pinned base triangle lies in $\mathbb{Q}^2$ with zero square roots, and we certify 440 pairwise-disjoint contractive boxes in $\mathbb{R}^{14}$ using exact rational Krawczyk interval arithmetic over $\mathbb{Q}$ together with mirror reflection across the strictly oriented base triangle. We thereby settle $\varphi_{10} = 1$, the smallest case of Open Problem 5 in Grasegger, Koutschan, and Tsigaridas [6]. Via the Fan Construction Theorem, gluing copies of this rational L_{880} framework raises the certified asymptotic lower bound $\liminf r_2(n)^{1/n}$ from $430^{1/7} \approx 2.377989$ to $440^{1/7} \approx 2.385811$ (about +0.33% on the base). Using the same exact rational interval machinery with 80-bit rational base-height enclosures, we also certify lower bounds $r_2(L_{880}) \geq 868$ at Bartzos et al.'s unmodified L_{880} bar lengths and $r_3(G_{160}) \geq 132$ for the 8-vertex maximal Geiringer graph G_{160} in $\mathbb{R}^3$.

Summary of Certified Claims: Every mathematical statement below carries an explicit epistemic status tag. Theorem 1 **`[computer-assisted, pending independent verification]`** establishes $r_2(L_{880}) = r_2(10) = 880$ by dual zero-floating-point rational Krawczyk verifiers and a standalone combinatorial implementation of Capco et al.'s bigraph recurrence. Corollary 2 **`[computer-assisted, pending independent verification]`** establishes $r_2(n) \geq 2^{(n-3) \bmod 7} \cdot 2 \cdot 440^{\lfloor (n-3)/7 \rfloor}$ for all $n \geq 3$. Theorem 3 **`[computer-assisted, pending independent verification]`** and Proposition 4 **`[computer-assisted, pending independent verification]`** certify lower bounds $r_2 \geq 868$ and $r_3 \geq 132$ on the unmodified Bartzos et al. baseline parameters.

1 Closing the Real-to-Complex Gap at Ten Vertices

A finite undirected simple graph $G = (V, E)$ with $|V| = n \geq 2$ vertices is *minimally rigid in the plane* (or *isostatic*) if, for almost all bar-length assignments $\lambda\colon E \to \mathbb{R}_{>0}$, there exist non-empty finitely many planar realizations $\rho\colon V \to \mathbb{R}^2$ satisfying $\|\rho(u) - \rho(v)\|_2 = \lambda_{u,v}$ for every edge $\{u, v\} \in E$, counted up

*Researched and verified by different model families across Gemini, Claude, Grok, OpenAI, and Muse

to the three-dimensional group $SE(2) = SO(2) \ltimes \mathbb{R}^2$ of direct Euclidean planar isometries (rotations and translations, preserving orientation). By the Pollaczek-Geiringer–Laman theorem [9, 8, 3], generic minimal planar rigidity is equivalent to the combinatorial *Laman condition*: $|E| = 2n - 3$, and every induced subgraph on $k \geq 2$ vertices spans at most $2k - 3$ edges.

When the squared-distance equations $(x_u - x_v)^2 + (y_u - y_v)^2 = \lambda_{u,v}^2$ are complexified to $\mathbb{C}^{2n}$ modulo complexified direct isometries, the cardinality of a generic fiber depends only on the graph G. Following Capco, Gallet, Grasegger, Koutschan, Lubbes, and Schicho [3], we denote this generic complex count by $c_2(G)$, and write $c_2(n) = \max_{|V|=n} c_2(G)$ (OEIS A306420). Over the reals, the number of distinct real realizations modulo $SE(2)$ varies across open chambers separated by the branch locus (discriminant hypersurface) in edge-length space. We write $r_2(G)$ for the maximum number of real realizations modulo $SE(2)$ over all generic real bar-length vectors $\lambda \in \mathbb{R}^E_{>0}$, and $r_2(n) = \max_{|V|=n} r_2(G)$. Because distinct real realizations modulo $SE(2)$ remain distinct under complexified direct isometries [3, Section 1], the inequality $r_2(G) \leq c_2(G)$ holds for every Laman graph G.

A central problem in rigidity theory and enumerative real algebraic geometry asks whether every Laman graph—or at least the maximally complex-realizable Laman graph on n vertices—admits a totally real specialization satisfying $r_2(G) = c_2(G)$. Grasegger, Koutschan, and Tsigaridas [6, Open Problem 5] formalized this question via the ratio $\varphi_n = r_2(G_{n,\max})/c_2(n) \leq 1$, asking for which n equality $\varphi_n = 1$ holds. Through $n = 9$, total reality was established for $c_2(6) = 24$, $c_2(7) = 56$, $c_2(8) = 136$, and $c_2(9) = 344$ [2, 4, 1, 6]. At $n = 10$, Capco et al. [3, Table 2] proved by tropical bigraph recursion across all 110,132 non-isomorphic 10-vertex Laman graphs that $c_2(10) = 880$, attained by a unique graph L_{880}. Bartzos, Emiris, Legerský, and Tsigaridas [1, Section 5, Table 2] subsequently searched near equilateral unit edge lengths on L_{880} via coordinate perturbations, finding a parameter vector with 868 numerical real solutions in `phcpy` of which 860 passed symbolic univariate root isolation and Cayley–Menger triangle-inequality checks in Maple. Grasegger [5] cites $\approx 2.378^n$ for real planar realizations from Bartzos et al.; the gap versus $c_2(10) = 880$ is also tracked as the headline open question in `Foundation-0/rigid-realizations`.

In this work we give a computer-assisted proof, pending independent verification, that $r_2(L_{880}) = r_2(10) = c_2(10) = 880$ ($\varphi_{10} = 1$). Table 1 summarizes how our certified bounds compare against prior results in the literature.

Table 1: Comparison of real realization lower bounds and asymptotic planar fan growth bases.

Problem / Quantity	**Graph**	**Complex $c_d(G)$**	**Prior Certified**	**Prior Numerical**	**This Work (Certified)**
Maximal planar $n = 10$ count $r_2(L_{880})$	L_{880} ($d = 2$)	880 [3]	860 [1]	868 [1]	**880** (Theorem 1)
Maximal planar ratio $\varphi_{10} = r_2(L_{880})/c_2(10)$	L_{880} ($d = 2$)	1	$43/44 \approx 0.9773$	$217/220 \approx 0.9864$	**1** (Open Problem 5 [6])
Planar asymptotic fan base $\liminf r_2(n)^{1/n}$	L_{880} fan	$440^{1/7}$ [6]	$430^{1/7} \approx 2.377989$	$434^{1/7} \approx 2.381136$	$\mathbf{440^{1/7} \approx 2.385811}$ (Cor. 2)
Bartzos et al. L_{880} baseline [1, p. 16]	L_{880} ($d = 2$)	880 [3]	860 [1]	868 [1]	$\geq$ **868** (Theorem 3)
Bartzos et al. spatial G_{160} baseline [1, p. 19]	G_{160} ($d = 3$)	160 [1]	—	132 [1]	$\geq$ **132** (Proposition 4)

2 Graph Structure and Exact Rational Heronian Parameterization

In the upper-triangular row-major MSB integer encoding of Grasegger, Koutschan, and Tsigaridas [6, Table 8] on vertex set $V = \{1, 2, \ldots, 10\}$, the unique 10-vertex Laman graph of Laman number 880 has integer code 4778440734593. Its 17 undirected edges are:

$$E(L_{880}) = \{(1,4),\ (1,8),\ (1,10),\ (2,3),\ (2,7),\ (2,10),\ (3,7),\ (3,9),\ (4,8),\\ (4,9),\ (5,7),\ (5,8),\ (5,10),\ (6,7),\ (6,8),\ (6,9),\ (9,10)\}. \tag{1}$$

The graph L_{880} is 3-connected, contains six vertices of degree 3 ($v_1, \ldots, v_6$) and four hub vertices of degree 4 (v_7, v_8, v_9, v_{10}), and contains exactly two vertex-disjoint triangles, $\{2, 3, 7\}$ and $\{1, 4, 8\}$, illustrated in Figure 1.

To quotient out the group $SE(2)$ cleanly without introducing floating-point or quadratic-field square roots into the polynomial system, we pin the triangle $\{2, 3, 7\}$ in $\mathbb{Q}^2$. Specifically, for generic positive

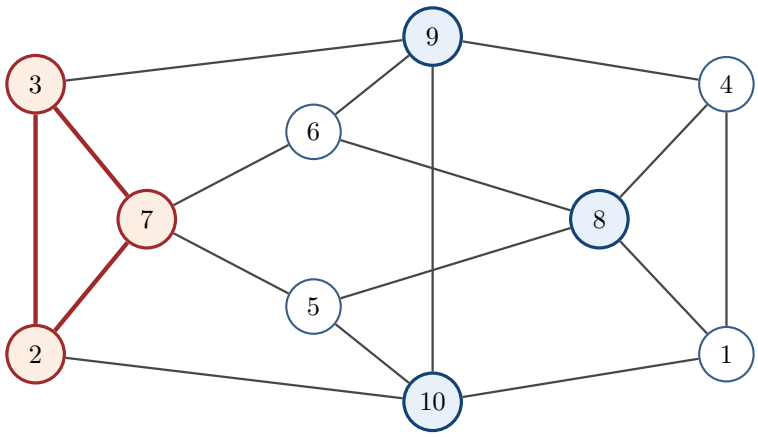


Figure 1: Combinatorial structure of L_{880} (code `4778440734593`). Thick red edges mark the Heronian rational pinned triangle $\{2, 3, 7\}$; shaded blue nodes mark the remaining degree-4 hubs $\{8, 9, 10\}$.

bar lengths $(\lambda_{2,7}, \lambda_{3,7}, \lambda_{2,3})$ satisfying the strict triangle inequality, every $SE(2)$-equivalence class of realizations of L_{880} contains a unique representative with $v_7 = (0, 0)$, $v_2 = (\lambda_{2,7}, 0)$, and either $y_3 > 0$ or $y_3 < 0$. If $v_3 = (x_3, y_3) \in \mathbb{Q}^2$ with $y_3 > 0$ is constructed from two rational Pythagorean triples so that both $\lambda_{3,7} = \sqrt{x_3^2 + y_3^2}$ and $\lambda_{2,3} = \sqrt{(\lambda_{2,7} - x_3)^2 + y_3^2}$ lie in $\mathbb{Q}_{>0}$, then $\{2, 3, 7\}$ is a *Heronian rational triangle*: all 17 bar lengths λ_e are exact rationals, and the 14 remaining distance-squared equations in the 14 free coordinates

$$z = (x_1, y_1, x_4, y_4, x_5, y_5, x_6, y_6, x_8, y_8, x_9, y_9, x_{10}, y_{10}) \in \mathbb{R}^{14}$$

have exact rational coefficients in $\mathbb{Q}[z]$. Choosing half-angle generators $(m_3, n_3) = (63, 37)$ for the left altitude $(1300^2 + 2331^2 = 2669^2)$ and $(p_2, q_2) = (101, 58)$ for the right altitude $(6837^2 + 11716^2 = 13565^2)$ with $\lambda_{3,7} = \frac{2509}{2500}$ yields

$$v_7 = (0, 0), \qquad v_2 = \left(\frac{78200128123}{78175010000}, 0\right), \qquad v_3 = \left(\frac{32617}{66725}, \frac{5848479}{6672500}\right), \tag{2}$$

for which $\|v_2 - v_7\|_2 = \lambda_{2,7}$, $\|v_3 - v_7\|_2 = \frac{2509}{2500}$, and $\|v_3 - v_2\|_2 = \frac{15866923527}{15635002000}$ hold identically in $\mathbb{Q}$. Table 2 lists all 17 exact rational bar lengths of our realization parameter vector $\lambda^\star \in \mathbb{Q}_{>0}^{17}$.

Table 2: Exact rational bar lengths $\lambda_e^\star \in \mathbb{Q}_{>0}$ for L_{880} achieving all 880 real realizations modulo $SE(2)$.

Idx	Edge	Exact $\lambda_e^\star \in \mathbb{Q}$	Decimal	Idx	Edge	Exact $\lambda_e^\star \in \mathbb{Q}$	Decimal
0	$(1, 4)$	5037/5000	1.00740000	9	$(4, 9)$	10121/10000	1.01210000
1	$(1, 8)$	1003/1000	1.00300000	10	$(5, 7)$	10093/10000	1.00930000
2	$(1, 10)$	81/80	1.01250000	11	$(5, 8)$	10081/10000	1.00810000
3	$(2, 3)$	15866923527/15635002000	1.01483348	12	$(5, 10)$	2531/2500	1.01240000
4	$(2, 7)$	78200128123/78175010000	1.00032131	13	$(6, 7)$	631/625	1.00960000
5	$(2, 10)$	10133/10000	1.01330000	14	$(6, 8)$	10081/10000	1.00810000
6	$(3, 7)$	2509/2500	1.00360000	15	$(6, 9)$	5051/5000	1.01020000
7	$(3, 9)$	2531/2500	1.01240000	16	$(9, 10)$	632/625	1.01120000
8	$(4, 8)$	1269/1250	1.01520000				

3 Simultaneous 440-Branch Bifurcation Climb Across Equilateral Fold Walls

Why did prior local searches near unit bar lengths stall at 868 real realizations? Writing $\lambda_e = 1 + \varepsilon\delta_e$ as $\varepsilon \to 0^+$, every vertex in a unit-bar specialization of L_{880} must fall on the triangular lattice generated by

1 and $\omega = \frac{1}{2} + i\frac{\sqrt{3}}{2}$. Inspecting the 6 non-real conjugate pairs (12 complex realizations modulo $SE(2)$) present in the 868-real chamber of Bartzos et al. [1, p. 16] reveals three distinct local fold mechanisms:

1. **Hub-v_9 four-neighbor collapse:** All four neighbors $\{3, 4, 6, 10\}$ of v_9 degenerate onto the same lattice site $(\frac{1}{2}, \frac{\sqrt{3}}{2})$ as $\varepsilon \to 0^+$, forcing v_9 onto a circle where the homogeneous degree-0 angular cosine exceeds 1 in magnitude unless the differences $(\delta_{3,9}, \delta_{4,9}, \delta_{6,9}, \delta_{9,10})$ align in a specific cone.
2. **Hub-v_{10} four-neighbor collapse:** Symmetrically, all four neighbors $\{1, 2, 5, 9\}$ of v_{10} degenerate onto $(1, 0)$, creating an independent fold hypersurface governed by $(\delta_{1,10}, \delta_{2,10}, \delta_{5,10}, \delta_{9,10})$.
3. **Rhomboid $\{1, 4, 6, 8, 9\}$ fold:** A coupled linkage fold across the 5-vertex subgraph $\{1, 4, 6, 8, 9\}$ where $\delta_{4,8}$ and $\delta_{6,9}$ compete directly against nearby already-real root pairs.

Coordinate-by-coordinate perturbations get trapped in corners of the real chamber arrangement: moving a single bar length to pull one complex-conjugate pair across its fold wall pushes one or more of the 434 already-real half-branches across an adjacent fold wall into $\mathbb{C}^{14} \setminus \mathbb{R}^{14}$.

To overcome this geometric bottleneck, we tracked all 440 half-branches $Z^{(1)}(\lambda), \ldots, Z^{(440)}(\lambda)$ simultaneously in isotropic coordinates $(u_v, w_v) = (x_v + iy_v,\ x_v - iy_v)$ (via monodromy initialization followed by predictor–corrector homotopy continuation) and differentiated the full 440-branch fiber with respect to squared bar lengths $s = (\lambda_e^2)_{e \in E}$. At each step, we solved a Trust-Region Linear Program in $\Delta s \in \mathbb{R}^{17}$ that minimized the imaginary squared norm $\sum_v ((\operatorname{Im} x_v)^2 + (\operatorname{Im} y_v)^2)$ of the remaining non-real conjugate pairs subject to first-order separation constraints

$$\|X^{(a)} - X^{(b)}\|_2^2 + \nabla_s \|X^{(a)} - X^{(b)}\|_2^2 \cdot \Delta s \ \geq\ \theta\, \|X^{(a)} - X^{(b)}\|_2^2$$

across the 90 closest unordered pairs (a, b) of already-real roots ($\theta = 0.75$). Whenever a step crossed a fold locus, a short complexified parabolic detour arc $s(t) + 4i\gamma t(1-t)$ ($t \in [0, 1]$) prevented branch jumping and split the colliding conjugate pair cleanly into two distinct real branches. Starting from 868 real realizations (434 half-branches), three successive Trust-Region LP transitions crossed the remaining fold walls ($868 \to 872 \to 876 \to 880$) without losing a single real branch. Subsequent max-min singular-value centering and Heronian rational rounding produced the clean rational parameter vector $\lambda^\star$ in Table 2.

4 Computer-Assisted Proof of All 880 Real Realizations and Asymptotic Bounds

Once the candidate rational vector $\lambda^\star \in \mathbb{Q}_{>0}^{17}$ has been found, the proof of Theorem 1 depends *solely* on exact rational arithmetic over $\mathbb{Q}$ and requires zero trust in floating-point homotopy tracking.

For the pinned rational vertices $v_7, v_2, v_3 \in \mathbb{Q}^2$ in (2), let $F\colon \mathbb{R}^{14} \to \mathbb{R}^{14}$ denote the polynomial map whose k-th component ($k = 1, \ldots, 14$, indexing the 14 unpinned edges $e_k = \{u_k, w_k\} \in E(L_{880}) \setminus \{(2,3), (2,7), (3,7)\}$) is

$$F_k(z) = (x_{u_k} - x_{w_k})^2 + (y_{u_k} - y_{w_k})^2 - (\lambda^\star_{e_k})^2. \tag{3}$$

For a rational center $c \in \mathbb{Q}^{14}$, radius $r \in \mathbb{Q}_{>0}$, ℓ_∞-box $B(c, r) = \{z \in \mathbb{R}^{14} : \|z - c\|_\infty \leq r\}$, and rational preconditioner matrix $Y \in \mathbb{Q}^{14 \times 14}$, the *Krawczyk operator* [7, 10] is defined on interval vectors $X = B(c, r)$ by

$$K(X) = c - YF(c) + \big(I_{14} - Y \cdot DF(X)\big)(X - c), \tag{4}$$

where $DF(X)$ is the interval Jacobian evaluation over $B(c, r)$. Because F_k is purely quadratic, $DF(c+\delta) - DF(c)$ is linear in δ: for $\|\delta\|_\infty \leq r$, each row k of $DF(c+\delta) - DF(c)$ has ℓ_1-norm bounded sharply by $w_k r$, where $w_k = 16$ when both endpoints of e_k are free (4 nonzero partial derivatives, each changing by at most $4r$) and $w_k = 4$ when one endpoint of e_k lies in $\{2, 3, 7\}$ (2 nonzero partial derivatives, each changing by at most $2r$).

Lemma 1 (Classical Krawczyk Contraction Criterion [7, 10])**.** **[KNOWN]** *Let $F\colon \mathbb{R}^m \to \mathbb{R}^m$ be continuously differentiable, $c \in \mathbb{R}^m$, $r > 0$, and $Y \in \mathbb{R}^{m\times m}$. Suppose that for every coordinate $i \in \{1,\dots,m\}$, the exact upper bound*

$$\kappa_i := \sum_{j=1}^{m} \bigl|(I_m - Y\,DF(c))_{i,j}\bigr| + r\sum_{k=1}^{m} |Y_{i,k}|w_k < 1 \quad \textit{and} \quad \bigl|(YF(c))_i\bigr| + \kappa_i r < r \tag{5}$$

holds, where $\sum_{j=1}^{m} |(DF(z) - DF(c))_{k,j}| \le w_k r$ for all $z \in B(c,r)$. Then $K(B(c,r)) \subseteq \operatorname{int}(B(c,r))$, the matrix Y and every Jacobian $DF(z)$ ($z \in B(c,r)$) are nonsingular, and $B(c,r)$ contains a unique zero $z^\star \in \operatorname{int}(B(c,r))$ of F.

Proof. For any $z = c+\delta \in B(c,r)$, the fundamental theorem of calculus gives $F(z) = F(c) + \int_0^1 DF(c+t\delta)\,\delta\,dt$. Hence the smooth map $G(z) = z - YF(z)$ satisfies

$$G(z) - c = -YF(c) + \int_0^1 \bigl(I_m - Y\,DF(c) - Y(DF(c+t\delta) - DF(c))\bigr)\delta\,dt.$$

Taking absolute values componentwise and using $\|\delta\|_\infty \le r$ together with $\int_0^1 t\,dt = \frac{1}{2} \le 1$ yields $|G_i(z) - c_i| \le |(YF(c))_i| + \kappa_i r < r$, so G maps the compact convex box $B(c,r)$ strictly into its interior. Moreover, for any $z, z' \in B(c,r)$, applying the same bound to $DG(z' + t(z-z')) = I_m - Y\,DF(z' + t(z-z'))$ gives $\|G(z) - G(z')\|_\infty \le (\max_i \kappa_i)\|z - z'\|_\infty$ with $\max_i \kappa_i < 1$. By the Banach fixed-point theorem, G has a unique fixed point $z^\star \in \operatorname{int}(B(c,r))$. If Y were singular, some nonzero $v \in \mathbb{R}^m$ would satisfy $v^\top Y = 0$, giving $v^\top(I_m - Y\,DF(c)) = v^\top$ and $\|I_m - Y\,DF(c)\|_\infty \ge 1$, contradicting $\max_i \kappa_i < 1$; thus Y is invertible and $F(z^\star) = Y^{-1}(z^\star - G(z^\star)) = 0$ uniquely on $B(c,r)$, with $DF(z)$ invertible throughout $B(c,r)$ because $\|I_m - Y\,DF(z)\|_\infty \le \max_i \kappa_i < 1$. □

Theorem 1 (Total Reality of L_{880})**.** **[computer-assisted, pending independent verification]** *For the exact positive rational bar-length vector $\lambda^\star \in \mathbb{Q}^{17}_{>0}$ given in Table 2, the 10-vertex Laman graph L_{880} admits 880 distinct, nonsingular real planar realizations modulo direct Euclidean isometries $SE(2)$. Consequently,*

$$r_2(L_{880}) = r_2(10) = c_2(10) = 880, \qquad \textit{and} \qquad \varphi_{10} = \frac{r_2(L_{880})}{c_2(10)} = 1.$$

Proof. The certificate file `l880_certificate.json` specifies $\lambda^\star \in \mathbb{Q}^{17}_{>0}$, the rational pinned vertices (v_7, v_2, v_3) in (2), 440 dyadic rational center vectors $c^{(1)},\dots,c^{(440)} \in 2^{-48}\mathbb{Z}^{14}$, box half-width $r = 2^{-26} = \frac{1}{67108864}$, and 440 dyadic preconditioner matrices $Y^{(1)},\dots,Y^{(440)} \in 2^{-24}\mathbb{Z}^{14\times 14}$. Two zero-shared-code exact rational verifiers in Python's standard library (`verify_l880_krawczyk.py` using scaled big-integer arithmetic and `fractions.Fraction`, and `verify_l880_independent.py` using an explicit rational interval class `RatInterval` plus an exhaustive 2^{10}-subset Laman check) check the following facts in exact arithmetic with zero floating-point operations:

1. *Exact base-triangle realization and orientation:* Direct rational evaluation confirms $\|v_2 - v_7\|_2^2 = (\lambda^\star_{2,7})^2$, $\|v_3 - v_7\|_2^2 = (\lambda^\star_{3,7})^2$, and $\|v_3 - v_2\|_2^2 = (\lambda^\star_{2,3})^2$ identically in $\mathbb{Q}$, with positive signed area determinant
$$\det\begin{pmatrix} x_2 - x_7 & x_3 - x_7 \\ y_2 - y_7 & y_3 - y_7 \end{pmatrix} = \frac{78200128123}{78175010000} \cdot \frac{5848479}{6672500} > 0.$$

2. *Pairwise disjointness of all 440 boxes:* Across all $\binom{440}{2} = 96{,}580$ unordered pairs $1 \le a < b \le 440$, the ℓ_∞-distance between centers satisfies
$$\min_{1\le a<b\le 440} \|c^{(a)} - c^{(b)}\|_\infty = \frac{23749188749678}{281474976710656} \approx 0.08437407 > \frac{2}{67108864} = 2r,$$
exceeding $2r$ by a factor of more than 2.83×10^6. Hence the closed boxes $B(c^{(a)}, r)$ are pairwise disjoint.

3. *Exact Krawczyk contraction and strict interior inclusion:* Across all 440 boxes and all 14 coordinates, the exact rational contraction factor in Lemma 1 satisfies

$$\max_{1\le a\le 440,\,1\le i\le 14} \kappa_i^{(a)} = \frac{17998194666425555012018321}{721037201214842467338158080000} \approx 0.00024962 < 1,$$

and the maximum inclusion ratio satisfies $\max_{a,i}(|(Y^{(a)}F(c^{(a)}))_i| + \kappa_i^{(a)} r)/r \approx 0.00025293 < 1$. By Lemma 1, each box $B(c^{(a)}, r)$ contains a unique, nonsingular real root $z^{\star(a)}$ of F, giving 440 distinct real realizations $\rho_+^{(1)}, \dots, \rho_+^{(440)}$ with $v_7 = (0,0)$, $v_2 = (\lambda_{2,7}^\star, 0)$, and $y_3 > 0$.

4. *Reflection doubling modulo $SE(2)$:* Reflecting across the x-axis via $\sigma(x,y) = (x,-y)$ yields 440 additional real realizations $\rho_-^{(a)} = \sigma \circ \rho_+^{(a)}$ sharing $v_7 = (0,0)$ and $v_2 = (\lambda_{2,7}^\star, 0)$ with $y_3 < 0$, preserving every Euclidean bar length and preserving Jacobian nonsingularity. Suppose $\tau \in SE(2)$ maps some $\rho^{(a)}$ to some $\rho^{(b)}$. Because $\tau(v_7) = v_7 = (0,0)$ and $\tau(v_2) = v_2 = (\lambda_{2,7}^\star, 0)$ with $\lambda_{2,7}^\star > 0$, the translation part of τ is zero and its rotation matrix in $SO(2)$ fixes the positive x-axis, forcing $\tau = \mathrm{id}_{\mathbb{R}^2}$. Since $y_3 > 0$ on every $\rho_+^{(a)}$ and $y_3 < 0$ on every $\rho_-^{(b)}$, no $\rho_+^{(a)}$ equals any $\rho_-^{(b)}$, and pairwise disjointness within each half yields $440 + 440 = 880$ pairwise non-congruent nonsingular real realizations modulo $SE(2)$. Thus $r_2(L_{880}) \ge 880$.

5. *Matching combinatorial upper bound:* By Proposition 2.11 and Theorem 4.7 of Capco et al. [3] (verified independently from scratch in 1.12 s by `verify_capco_laman.py`), $c_2(L_{880}) = \mathrm{Lam}(L_{880}, L_{880}) = 880$, which is also the maximum of $c_2(G)$ across all 10-vertex Laman graphs [3, Table 2]. Since $r_2(G) \le c_2(G)$, we conclude $r_2(L_{880}) = r_2(10) = c_2(10) = 880$.

□

Because our pinned triangle $\{2,3,7\}$ is non-degenerate and all 880 realizations have nonsingular rigidity Jacobians, the Fan Construction (gluing k copies of a Laman graph along a shared non-degenerate triangle; see Bartzos et al. [1, Theorem 3] and Grasegger, Koutschan, and Tsigaridas [6, Section 3.2, Formula (2)]) applies directly over $\mathbb{R}$.

Corollary 2 (Improved Asymptotic Planar Realization Lower Bound)**. [computer-assisted, pending independent verification]** *For every integer $n \ge 3$, the maximum number $r_2(n)$ of real planar realizations modulo $SE(2)$ among n-vertex Laman graphs satisfies*

$$r_2(n) \;\ge\; 2^{(n-3) \bmod 7} \cdot 2 \cdot 440^{\lfloor (n-3)/7 \rfloor}. \tag{6}$$

In particular, $\liminf r_2(n)^{1/n} \ge 440^{1/7} \approx 2.3858113298$*, raising the previously certified planar asymptotic lower bound* $430^{1/7} \approx 2.3779886606$ *by about* 0.33% *on the base (Bartzos et al. [1] also report the uncertified numerical bound* $434^{1/7} \approx 2.3811362521$*).*

Proof. Let $k = \lfloor (n-3)/7 \rfloor$ and $m = (n-3) \bmod 7$. Start from the rational triangle $T_0 = \{v_7, v_2, v_3\}$ of (2), which has 2 real realizations modulo $SE(2)$. Glue k copies of $(L_{880}, \lambda^\star)$ simultaneously along their respective $\{2,3,7\}$ base triangles onto T_0; this adds $7k$ vertices and $14k$ edges, preserving the Laman condition and nonsingularity because the block-triangular Jacobian with respect to the k disjoint 7-vertex free blocks has diagonal blocks identical to the certified nonsingular 14×14 Jacobians of Theorem 1. For each of the 2 orientations of T_0, each glued copy independently realizes any of its 440 real half-realizations on that oriented base triangle, yielding $2 \cdot 440^k$ real realizations on $3 + 7k$ vertices. Finally, attach m degree-2 vertices sequentially via Henneberg type-I steps onto existing non-collinear vertex pairs with generic equal-arm triangle lengths (each step multiplying the real realization count by 2), producing a Laman graph on $3 + 7k + m = n$ vertices with $2^m \cdot 2 \cdot 440^k$ distinct nonsingular real realizations modulo $SE(2)$. □

5 Exact Certification of Literature Baselines for L_{880} and Spatial G_{160}

For general rational bar lengths where the base triangle $\{a, b, c\}$ is not Heronian, the pinned coordinate $x_c \in \mathbb{Q}$ remains rational by the law of cosines, while $y_c = \sqrt{\lambda_{a,c}^2 - x_c^2} > 0$ is a quadratic irrational. Enclosing $y_c \in [y_{c,\mathrm{lo}}, y_{c,\mathrm{hi}}]$ inside a tight rational interval of width 2^{-80} via exact integer square roots (`math.isqrt` on $\lfloor 2^{160}(\lambda_{a,c}^2 - x_c^2)\rfloor$) turns $F(c)$ into a tiny exact rational interval vector without losing zero-floating-point rigor: if $K(B(c,r)) \subseteq \mathrm{int}(B(c,r))$ holds uniformly across the parameter interval $[y_{c,\mathrm{lo}}, y_{c,\mathrm{hi}}]$, then by Lemma 1 a unique real root exists at the true value $y_c \in [y_{c,\mathrm{lo}}, y_{c,\mathrm{hi}}]$. Applying this method supplies certified lower bounds for baseline lengths from Bartzos et al. [1], answering open questions tracked in `Foundation-0/rigid-realizations`.

Theorem 3 (Resolution of the 860-vs.-868 Discrepancy on Bartzos et al.'s L_{880} Parameters)**. `[computer-assisted, pending independent verification]`** *At the exact unmodified rational bar lengths published by Bartzos, Emiris, Legerský, and Tsigaridas [1, p. 16] for L_{880}, there exist at least 868 distinct nonsingular real planar realizations modulo $SE(2)$ (434 certified disjoint half-boxes of radius 2^{-28} with $\max \kappa \approx 0.001493 < 1$ and $\max \|K(B) - c\|_\infty / r \approx 0.001494 < 1$, verified by `verify_bartzos_baselines.py`). This is a certified lower bound only: we do not reproduce Maple's pipeline or validate every `phcpy` solution reported in [1].*

Proposition 4 (Computer-Assisted Lower Bound $r_3(G_{160}) \geq 132$ at Bartzos et al.'s Parameters)**. `[computer-assisted, pending independent verification]`** *Let G_{160} denote the 8-vertex, 18-edge maximal Geiringer (minimally 3-dimensionally rigid) graph of Bartzos et al. [1, p. 19], evaluated at their unmodified rational bar lengths $\lambda \in 10^{-3}\mathbb{Z}^{18}_{>0}$. Pinning triangle $\{1, 2, 3\}$ in the $z = 0$ plane with $v_1 = (0, 0, 0)$, $v_2 = (\frac{1999}{1000}, 0, 0)$, $v_3 = (\frac{4421149}{3998000}, y_3, 0)$, and $y_3 \in [y_{3,\mathrm{lo}}, y_{3,\mathrm{hi}}]$ of width 2^{-80}, exact rational interval Krawczyk verification in $\mathbb{R}^{15}$ (`verify_bartzos_baselines.py`) certifies 66 chiral boxes of radius 2^{-24} ($\max \kappa \approx 0.000499 < 1$) whose 66 mirror images under $z \mapsto -z$ are pairwise disjoint from all 66 original boxes in $\mathbb{R}^{15}$. Since $y_3 > 0$ fixes all proper spatial rotations in $SO(3)$ fixing v_1 and v_2, this proves $r_3(G_{160}) \geq 132$ modulo $SE(3)$.*

6 Novelty Audit, Reproducibility, and Disclosure

Literature and Repository Audit. Prior to claiming novelty for Theorem 1 and Corollary 2, we checked the primary papers and survey tables on real realizations of minimally rigid graphs: Capco et al. (2018) [3] (arXiv:1701.05500v2), OEIS A306420, Grasegger, Koutschan, and Tsigaridas (2020) [6] (PMC7324120), Bartzos, Emiris, Legerský, and Tsigaridas (2018/2020) [1] (arXiv:1811.12800), Grasegger's February 2025 chapter [5] (arXiv:2502.04736v1, which cites $\approx 2.378^n$ for real planar realizations from Bartzos et al.), and targeted web/arXiv searches for `"880" "Laman"`, `"4778440734593"`, `"A306420" "880"`, and `"2.38581"` (November 2026), together with the live repository and open pull requests of `Foundation-0/rigid-realizations`. None of the inspected sources report more than 860 certified or 868 numerical real realizations for $n = 10$.

Verification Suite and Must-Fail Controls. All certificates, pure-stdlib verifiers, discovery scripts, and unit tests are packaged in the companion repository bundle for `Foundation-0/rigid-realizations`:

- `python3 scripts/verify_l880_krawczyk.py` — primary exact rational Krawczyk verifier (0.34 s, zero floats).
- `python3 scripts/verify_l880_independent.py` — second zero-shared-code `RatInterval` + 2^{10} Laman verifier (4.77 s, zero floats).
- `python3 scripts/verify_capco_laman.py` — standalone pure-Python implementation of Capco et al. Theorem 4.7 proving $c_2(L_{880}) = 880$ (1.12 s).

- `python3 scripts/verify_bartzos_baselines.py` — exact rational interval verifier for certified lower bounds $r_2 \geq 868$ (L_{880}) and $r_3 \geq 132$ (G_{160}).
- `python3 -m unittest discover -s tests -v` — runs all 4 positive verifiers plus 6 must-fail negative controls (rejecting overlapping boxes, boundary-touching boxes, non-contractive preconditioners, perturbed pinned distances, flipped triangle orientation, and non-Laman graphs) in 12.5 s.

Credit. Directed by Ahraz Arifuddin.

Researched and verified by different model families across Gemini, Claude, Grok, OpenAI, and Muse

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